

# Supplementary Material for Dependency-Frontier Scheduling with Asymmetric-Cost Spill Planning for NPU Kernels

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## Contents

|          |                                                                             |           |
|----------|-----------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Problem Definition and Evaluator Semantics</b>                           | <b>2</b>  |
| 1.1      | Input DAG and buffer lifetimes . . . . .                                    | 2         |
| 1.2      | P1: logical-lifetime peaks . . . . .                                        | 2         |
| 1.3      | P2: contiguous addresses and explicit spill artifacts . . . . .             | 2         |
| 1.4      | Static classes and the P2 objective . . . . .                               | 3         |
| 1.5      | P3: pipelined time on the augmented DAG . . . . .                           | 3         |
| 1.6      | Validation, reproduction, and data scope . . . . .                          | 3         |
| <b>2</b> | <b>Formal Results and Their Boundaries</b>                                  | <b>3</b>  |
| 2.1      | Canonical traffic decomposition and the factor-two bound . . . . .          | 4         |
| 2.2      | Fixed-order residency-gap model . . . . .                                   | 4         |
| 2.3      | From a lower bound to a machine-checkable certificate . . . . .             | 5         |
| 2.4      | Candidate-portfolio monotonicity . . . . .                                  | 5         |
| 2.5      | Complexity and exact-backend boundary . . . . .                             | 6         |
| <b>3</b> | <b>Solver Implementation and Complete Walkthrough</b>                       | <b>6</b>  |
| 3.1      | Canonical scalable solver . . . . .                                         | 6         |
| 3.2      | A step-by-step reproducible small graph . . . . .                           | 8         |
| 3.3      | End-to-end path on real Conv_Case0 . . . . .                                | 9         |
| 3.4      | Offline cost repair and the fixed-order oracle . . . . .                    | 9         |
| 3.5      | Failures, unfinished cases, and reproduction boundary . . . . .             | 9         |
| <b>4</b> | <b>Complete Evidence and Negative Results</b>                               | <b>10</b> |
| 4.1      | Complete public P1, P2, and P3 results . . . . .                            | 10        |
| 4.2      | Selected ablation path and descriptive nonadditivity . . . . .              | 11        |
| 4.3      | Three spill-cost regimes under $Tr=Vol+Dt$ . . . . .                        | 12        |
| 4.4      | Within-portfolio P2-to-P3 tradeoff . . . . .                                | 13        |
| 4.5      | Case-study repair and fixed-order research oracle . . . . .                 | 13        |
| 4.6      | Single-instance capacity boundary . . . . .                                 | 15        |
| 4.7      | Standard validated re-evaluation on internal synthetic benchmarks . . . . . | 17        |

## Supplement Overview

This supplement gives auditable details for the paper’s exact-to-heuristic evidence chain. Section 1 specifies P1/P2/P3 evaluator semantics and static backed/generated classes. Section 2 proves the traffic identity, fixed-order lower bound, and certificate, with a contiguous-address counterexample. Section 3 uses a pedagogical graph and Conv\_Case0 for an end-to-end walkthrough while separating production, nonuniform repair studies, and the fixed-order oracle. Section 4 reports complete P1/P2/P3 results, component attribution, one capacity sweep, all 36 synthetic and eight oracle instances, artifact boundaries, and negative results.

## 1 Problem Definition and Evaluator Semantics

This section states the complete contract among the input, output, and evaluator. The distinction matters: P1 logical-lifetime peaks, P2 physical residency segments, and P3 pipelined completion time are different objects and cannot substitute for one another. Likewise, the evaluator's backed/unbacked classification is not a dynamic hardware dirty bit.

### 1.1 Input DAG and buffer lifetimes

The input is a directed acyclic graph  $G = (N, E)$  with three node classes:

1. an ALLOC node creates a buffer;
2. a FREE node ends that buffer's logical lifetime; and
3. every other node is a compute or transfer operation listing the buffers it accesses, its execution pipeline, and its cycle count.

Each buffer  $b$  has a unique identifier, size  $s_b$ , memory type  $\tau(b) \in \{L1, UB, L0A, L0B, L0C\}$ , a unique allocation node  $a_b$ , and a unique free node  $f_b$ . We call the occurrences of  $b$  at operation nodes, together with  $a_b$  and  $f_b$ , its *mandatory events*. An original schedule  $\pi = (\pi_0, \dots, \pi_{|N|-1})$  must contain every original node exactly once and satisfy  $\text{pos}_\pi(u) < \text{pos}_\pi(v)$  for every  $(u, v) \in E$ .

Two boundaries are easy to miss. First, the data record only which buffers an operation accesses; it does not expose per-access read/write roles. Second, the canonical evaluator treats FREE as a mandatory-residency event: a previously evicted buffer must be reloaded before it can be freed.

### 1.2 P1: logical-lifetime peaks

P1 emits neither addresses nor spills. For each memory type, the evaluator scans the original schedule, adding the size at ALLOC (or incrementing a buffer count for an L0 type) and subtracting it at FREE. The L1/UB metrics are therefore maxima of ALLOC/FREE prefix sums; the L0A/L0B/L0C metrics are maximum counts of simultaneously logically live buffers.

This is a *logical-lifetime statistic*, not the bytes occupying physical addresses at a P2 instant. A logically live buffer may be temporarily evicted in P2. Conversely, the P1 peak contains neither address fragmentation, spill-out/spill-in nodes, nor address-reuse dependencies. P1 thus describes logical pressure induced by an order; by itself it does not prove that a legal contiguous-address layout exists.

### 1.3 P2: contiguous addresses and explicit spill artifacts

A P2 artifact has three parts: an augmented schedule  $\hat{\pi}$  containing synthetic spill-out/spill-in nodes, the initial residency address  $m_b$  of every buffer, and an ordered spill table of pairs  $(b, m'_b)$ . Spill-table entry  $j$  corresponds to two deterministically numbered synthetic nodes  $o_j$  (spill-out) and  $i_j$  (spill-in);  $m'_b$  is the address of the new residency segment after reload.

A P2 artifact is legal exactly when it passes all of the following checks:

1. **Coverage and topology:** every original and synthetic spill node occurs exactly once, and all original DAG dependencies hold.
2. **Spill order:**  $a_b \prec o_j \prec i_j \prec f_b$ ; no operation accesses  $b$  between  $o_j$  and  $i_j$ ; and two spill intervals of the same buffer do not overlap.
3. **Address range:** every residency segment occupies a half-open interval  $[m_b, m_b + s_b)$  or  $[m'_b, m'_b + s_b)$  wholly inside its memory capacity.
4. **Space-time nonoverlap:** residency segments of the same memory type whose lifetimes overlap in schedule position must have disjoint address intervals.

The fourth condition expresses both capacity and contiguity. Checking only that total bytes do not exceed capacity is insufficient: several small holes can have enough total space while no single hole fits a new contiguous

segment.

#### 1.4 Static classes and the P2 objective

The evaluator makes a *static* binary classification. A buffer occurring in any COPY-IN operation is input-backed (abbreviated backed/clean); every other buffer is generated and unbacked (generated/dirty). The classification never changes during execution and does not infer whether an access is a read or a write.

Interrupting a backed buffer requires only a reload of  $s_b$  bytes; interrupting an unbacked buffer requires a write-back and a reload, charging  $2s_b$ . Let  $C$  and  $D$  be the cumulative raw spill volumes of the two classes and let  $V = C + D$ . Extra traffic is

$$E = C + 2D = (C + D) + D = V + D. \quad (1)$$

P2 lexicographically minimizes  $K_2 = (E, n_{\text{spill}}, T)$ : extra traffic first, then the number of spill entries, then completion time. Every P2 comparison in the paper uses the matching official P2 artifact; values from official P3 artifacts are never mixed into that comparison.

#### 1.5 P3: pipelined time on the augmented DAG

P3 uses the same physical legality conditions as P2, but makes augmented-schedule completion time the primary objective. For each node, the evaluator takes the maximum of all predecessor completion times, the availability time of the node’s hardware pipeline, and the reuse-ready time of every address interval it occupies; it then adds the node’s cycle count. Synthetic spill-out/spill-in nodes and their dependencies participate in the simulation. A pipeline is serialized, and a new residency segment reusing an address range waits for the previous segment to finish. The objective key is  $K_3 = (T, E, n_{\text{spill}})$ .

P2 and P3 are therefore not two spellings of the same objective. A wider early-reload window may increase traffic yet hide transfers behind computation and reduce  $T$ ; P3 accepts that exchange when  $K_3$  improves. We describe only the observed bounded-portfolio traffic–time tradeoff, not a global Pareto frontier.

#### 1.6 Validation, reproduction, and data scope

We keep three levels distinct: a model objective, a generated physical artifact, and canonical-evaluator validation. A low relaxation objective is not a submission. We report an artifact as legal only after addresses, spill dependencies, and the schedule are generated and the evaluator returns zero violations. For the fixed-order exact backend, a JSON record together with matching schedule/memory/spill text files forms a replayable certificate; a metric-only ledger requires rerunning the deterministic solver to reconstruct the complete artifact.

Table S1 gives the original-node counts of the six public instances. The repository also contains 36 internal synthetic instances and eight 17-node fixed-order oracle instances. They support boundary and mechanism checks but are not additional public datasets. The solver has no case-name branch. However, the retained P1 fallback contains operation-type-specific tie-breaks, so the solver as a whole is not motif-agnostic; Section 3 states the precise boundary.

**Table S1:** Original DAG sizes of the six public instances. FA denotes FlashAttention.

| Instance     | Nodes | Instance     | Nodes  |
|--------------|-------|--------------|--------|
| Conv_Case0   | 2,580 | Conv_Case1   | 36,086 |
| FA_Case0     | 1,716 | FA_Case1     | 6,952  |
| Matmul_Case0 | 4,160 | Matmul_Case1 | 30,976 |

## 2 Formal Results and Their Boundaries

This section gives the complete claims supported by the current evidence. The proofs deliberately distinguish an accounting identity, a capacity-relaxation lower bound, a fixed-order certificate, and global optimality over topological orders; none may be substituted for another.

## 2.1 Canonical traffic decomposition and the factor-two bound

For residency interruption  $i$ , let  $v_i$  be the buffer size, and let  $d_i = v_i$  if the buffer is unbacked and  $d_i = 0$  otherwise. Define  $V = \sum_i v_i$  and  $D = \sum_i d_i$ .

**Proposition 1 (accounting identity and tight bounds).** Every legal P2/P3 artifact satisfies

$$E = V + D, \quad V \leq E \leq 2V. \quad (2)$$

At fixed  $V > 0$ , changing only the class composition of spilled bytes can change the traffic ratio between two artifacts by at most two, and this bound is attainable.

**Proof.** A backed interruption costs  $v_i$ , while an unbacked interruption costs  $2v_i = v_i + d_i$ . Summing term by term gives  $E = \sum_i (v_i + d_i) = V + D$ . Since  $d_i \in \{0, v_i\}$ ,  $0 \leq D \leq V$ , hence  $V \leq E \leq 2V$ . If all spilled bytes are backed,  $D = 0, E = V$ ; if all are unbacked,  $D = V, E = 2V$ . Both endpoints are attainable.  $\square$

Proposition 1 is an evaluator accounting property, not a causal mechanism claim. It neither proves that production candidate-order generation reads class labels (it does not) nor that  $D$  is more important than  $V$ . A mechanism comparison must report both  $\Delta V$  and  $\Delta D$ ; at fixed  $V$ , class composition alone cannot produce a gain greater than two.

## 2.2 Fixed-order residency-gap model

Fix a legal original topological order  $\pi$ . For each buffer  $b$ , sort its mandatory-event positions as  $p_{b,0} < p_{b,1} < \dots < p_{b,r_b}$ . The open interval between consecutive events defines a residency gap  $g = (b, j)$ . Binary variable  $x_g = 1$  keeps  $b$  continuously resident from event  $j$  to event  $j + 1$ ;  $x_g = 0$  permits an interruption. Let

$$w_g = \begin{cases} s_b, & b \text{ is input-backed,} \\ 2s_b, & b \text{ is unbacked.} \end{cases} \quad (3)$$

For memory type  $k$  and schedule cut  $t$ , let  $P_{k,t}$  be the bytes mandatorily pinned at that cut, let  $\mathcal{G}_{k,t}$  be the optional same-type gaps crossing it, and let capacity be  $B_k$ . The fixed-order weighted-gap model is

$$L^* = \min_x \sum_g w_g(1 - x_g), \quad (4)$$

$$\text{s.t. } P_{k,t} + \sum_{g \in \mathcal{G}_{k,t}} s_g x_g \leq B_k, \quad \forall k, t, \quad (5)$$

$$x_g \in \{0, 1\}.$$

If  $P_{k,t} > B_k$  at any cut, spilling cannot repair the fixed order because the mandatory pinned set of one event already exceeds capacity.

**Proposition 2 (fixed-order traffic lower bound).** The optimum  $L^*$  of Eqs. (4)–(5) is no greater than the traffic of any legal fixed-order plan with contiguous physical addresses.

**Proof.** Take any legal physical plan  $A$ . Set  $x_g = 1$  exactly when the buffer remains continuously resident throughout gap  $g$ , and set it to zero otherwise. At every cut, the mandatory residency segments and every selected continuous gap really do occupy the same memory type simultaneously. Because  $A$  is legal, their total size is at most  $B_k$ , so the constructed  $x$  satisfies every capacity constraint.

Every gap with  $x_g = 0$  contains at least one complete residency interruption, so  $A$  pays at least  $w_g$  there; an artificial plan could interrupt the same gap more than once and pay more. Thus  $E(A) \geq \sum_g w_g(1 - x_g) \geq L^*$ . The argument never uses the concrete addresses of  $A$ , so it holds for every legal contiguous-address plan.  $\square$

### 2.3 From a lower bound to a machine-checkable certificate

The capacity relaxation discards address-shape information, so its solution need not be packable even when all addresses may be globally reassigned from scratch. Consider a discrete storage-allocation counterexample with capacity four. Seven segments have “active cuts/size”

$$A : \{0\}/2, B : \{0, 1\}/2, C : \{1, 2\}/1, D : \{1, 2, 3\}/1, E : \{2, 3\}/1, F : \{3, 4\}/2, G : \{4\}/2.$$

The active volumes at the five cuts are 4, 4, 3, 4, 4, so every cumulative-capacity constraint holds, yet no set of fixed contiguous offsets exists. At cut 0,  $A$  and  $B$  must occupy the two length-two halves; without loss of generality take  $B = [0, 2)$ . At cut 1,  $C$  and  $D$  must occupy  $[2, 3)$  and  $[3, 4)$ . If  $D = [2, 3)$ , then  $E$  at cut 2 must occupy a unit position in  $[0, 2)$ ; at cut 3,  $D$  and  $E$  fragment the remaining space so the length-two  $F$  cannot fit. If  $D = [3, 4)$ , only  $E = [0, 1)$  permits  $F = [1, 3)$  at cut 3; then at cut 4 only one unit remains on each side of  $F$ , so length-two  $G$  cannot fit. Other  $E$  positions already fail at cut 3, and the left-right mirror is symmetric. The gap model accepts these intervals because it sees only volume; “gap optimal” alone therefore proves neither physical feasibility nor canonical validity.

**Theorem 3 (fixed-order traffic certificate).** For a given fixed order, suppose that

1. the gap model proves optimum  $L^*$ ;
2. the selected gaps are realized as contiguous-address residency segments and every spill-out/spill-in dependency is legal; and
3. the canonical evaluator returns zero violations and computes  $E(A) = L^*$ .

Then artifact  $A$  has minimum extra traffic for that fixed order.

**Proof.** By Proposition 2, every legal fixed-order physical plan  $A'$  satisfies  $E(A') \geq L^*$ . Conditions 2–3 make  $A$  a legal plan with  $E(A) = L^*$ , so no legal plan under the same order has lower traffic.  $\square$

The qualifiers in Theorem 3 are essential. “OPTIMAL” from greedy packing or NoOverlap2D means that its satisfaction model found or proved a layout; it is not a second layout-quality optimization. Packing failure or timeout does not invalidate  $L^*$  as a lower bound. The certificate also does not minimize spill count or time among traffic ties, so it is not complete P2 optimality; fixed-order traffic optimality is not global optimality over topological orders.

### 2.4 Candidate-portfolio monotonicity

Let  $\mathcal{A}$  be the artifacts successfully generated inside the candidate loop, and let  $K$  be the implementation’s P2- or P3-semantic lexicographic key. The production solver returns  $A^* = \arg \min_{A \in \mathcal{A}} K(A)$ .

**Proposition 4 (internal-key monotonicity under candidate expansion).** For any additional successfully generated candidate set  $\mathcal{B}$ ,

$$\min_{A \in \mathcal{A} \cup \mathcal{B}} K(A) \leq_{\text{lex}} \min_{A \in \mathcal{A}} K(A). \quad (6)$$

**Proof.**  $\mathcal{A} \subseteq \mathcal{A} \cup \mathcal{B}$ , so the previous best candidate remains in the expanded set; minimizing over the larger set cannot be worse.  $\square$

This is monotonicity of the portfolio’s *internal selection key*, not a causal claim about a new candidate or a legality theorem. Calling the return value a legal artifact additionally requires a correct generator or canonical evaluation of the returned artifact. The production loop does not invoke that evaluator on every candidate; our experiments report only final artifacts with zero violations. If the portfolio contains a strong baseline order, non-regression against it follows from set inclusion and does not attribute an improvement to ordering, placement, or a victim component. Production selection computes the actual  $K_2/K_3$ ; it needs no overflow-area surrogate theorem.

## 2.5 Complexity and exact-backend boundary

For fixed numbers of candidate orders, victim policies, and early-reload windows, the scalable path performs topological ordering, residency scans, address placement, and timing simulation. A placement request may scan the resident set or free intervals, giving a conservative quadratic term in the number of buffers; the portfolio contributes only a fixed multiplier. This is an implementation upper bound, not a linear-time guarantee.

Weighted-gap selection and fallback `NoOverlap2D` are exponential in the worst case. The CP-SAT backend is therefore an offline research oracle, not part of canonical `solve`. A future integration would need a size threshold or timeout and a validated fallback when optimization is unfinished, packing fails, or validation fails. Nonzero gaps, timeouts, and not-run cases in the current experiments must be reported as such and cannot be extrapolated into universal exactness.

## 3 Solver Implementation and Complete Walkthrough

This section first separates three solver roles and then expands how a DAG becomes a validated artifact. The roles are the default scalable portfolio, offline cost repair used only for mechanism studies, and a fixed-topological-order CP-SAT oracle. The latter two are not hidden stages of production `solve`.

### 3.1 Canonical scalable solver

For P2/P3, production generates four structural topological orders: `unlock-frontier`, `capfit-id`, the P1 memory order, and `id-raw`. Only the first is new in v2. The other three provide historical and structural diversity and are not relabeled as cost-aware order generators. None of the four reads backed/generated labels. Class information enters only the two victim scores and the true traffic computation and final selection over complete candidates. The phrase “purely structural” applies only to the new `unlock-frontier`: the retained P1 fallback contains D2S-, COPY-IN-, and MOVE-specific tie-breaks for certain input shapes. We therefore do not call the canonical solver as a whole motif-agnostic; the narrow valid statement is that frontier reads neither instance names, operation motifs, nor buffer classes.

`unlock-frontier` schedules ready FREE nodes first, then the minimum-ID ready operation. When only ALLOC nodes remain, it finds the minimum-ID consumer whose remaining predecessors are all ready and completes that predecessor group. This avoids a stream of single-input transfers indefinitely starving a multi-input operation.

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#### Algorithm 1 Dependency-frontier topological order

---

**Require:** DAG instance  $I$

```

1: Partition zero-indegree nodes into  $Q_f, Q_o, Q_a$ ;  $S \leftarrow []$ 
2: while  $Q_f \cup Q_o \cup Q_a \neq \emptyset$  do
3:   if  $Q_f \neq \emptyset$  then
4:      $v \leftarrow$  maximum-ID node in  $Q_f$ 
5:   else if  $Q_o \neq \emptyset$  then
6:      $v \leftarrow$  minimum-ID node in  $Q_o$ 
7:   else
8:      $u \leftarrow$  minimum-ID consumer whose unfinished predecessors are all in  $Q_a$ 
9:      $v \leftarrow$  minimum-ID ready predecessor of  $u$ 
10:    if no such  $u$  exists then
11:       $v \leftarrow$  minimum-ID node in  $Q_a$ 
12:    end if
13:  end if
14:  append  $v$  to  $S$ ; update indegrees and  $Q_f, Q_o, Q_a$ 
15: end while
16: return  $S$ 

```

---

Each topological order runs true best-fit address placement: among all sufficiently large free intervals, choose the shortest and then the lowest-address tie; merge adjacent intervals on release. If no contiguous hole fits, the placer chooses a victim not pinned by the next operation, inserts spill-out, and releases its entire address segment until placement succeeds. A later access or FREE inserts spill-in, which may place the new residency segment at a different address.

The production portfolio compares `dist-size-cost` and `share-adaptive-25`. The former combines next-use distance, size, and static traffic cost. The latter changes priority when backed bytes are scarce or when a generated-only memory type has highly heterogeneous sizes. P2 enumerates early-reload windows  $H \in \{0, 5, 40\}$ ; P3 uses  $H \in \{0, 5, 40, 80, 120\}$ . An early reload occurs only when a sufficiently large contiguous hole already exists and never evicts another resident, but it changes later residency state, so every window is evaluated as a complete candidate.

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**Algorithm 2** Bounded P2/P3 artifact portfolio

---

**Require:** Instance  $I$  and problem number  $q \in \{2, 3\}$

- 1:  $\mathcal{O} \leftarrow \{\text{frontier, capfit-id, P1, id-raw}\}$
- 2:  $\mathcal{P} \leftarrow \{\text{dist-size-cost, share-adaptive-25}\}$
- 3:  $\mathcal{H} \leftarrow \{0, 5, 40\}$ ; if  $q = 3$ , add 80, 120
- 4:  $A^* \leftarrow \emptyset$
- 5: **for**  $(O, P, H) \in \mathcal{O} \times \mathcal{P} \times \mathcal{H}$  **do**
- 6:   generate topological order  $\pi_O$
- 7:   generate addresses and spill artifact  $A$  using best-fit,  $P$ , and  $H$
- 8:   **if** placement fails **then**
- 9:     skip this infeasible portfolio member
- 10:   **end if**
- 11:   compute  $E, n_{\text{spill}}, T$  (the loop does not call the canonical evaluator)
- 12:   update  $A^*$  by  $K_2 = (E, n_{\text{spill}}, T)$  or  $K_3 = (T, E, n_{\text{spill}})$
- 13: **end for**
- 14: return  $A^*$ ; the experimental harness canonically evaluates the final artifact

---

Algorithm 2 considers at most  $4 \times 2 \times |\mathcal{H}|$  members and skips failed placements. It generates address/spill artifacts and computes the true objective for successful candidates; neither live area nor overflow area selects the winner. Production `solve` does not canonically evaluate every candidate inside the loop. The experimental harness validates the final selected artifact, and we report only zero-violation results. Table S2 lists deterministic details omitted from the main-paper pseudocode.

**Table S2:** Fixed constants and boundary rules of the canonical solver.

| Mechanism          | Value                      | Exact role                                                                                                                                          |
|--------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontier ties      | FREE max ID; operation min | The ALLOC key is (minimum unlockable consumer ID, ALLOC ID); its first component is infinity when no consumer is unlockable.                        |
| Next-use cap       | $2^{20}$                   | Cap base-order distance before either victim score is formed.                                                                                       |
| Size heterogeneity | $\max/\min \geq 8$         | If a memory type contains no backed buffer, adaptive prioritizes releasing a large contiguous segment; otherwise it retains distance priority.      |
| Backed share       | $\leq 25\%$                | When an evictable backed victim exists and backed bytes are at most this share of evictable resident bytes, use this low-cost reserve first.        |
| Early reload       | $\text{distance} \leq H$   | Use only an existing fitting hole; prefetch does not create a victim.                                                                               |
| Pinned relocation  | at most 100 per request    | Only under fragmentation with no unpinned victim, spill and immediately reload the smallest pinned buffer; exceeding the limit fails the candidate. |

### 3.2 A step-by-step reproducible small graph

Table S3 is a pedagogical example manually expanded according to the current best-fit/spill implementation and canonical-evaluator rules. It shows how spill nodes, addresses, and charging form a P2 artifact; it is not a run log, public benchmark, or oracle datum and supports no numerical benchmark claim. Number the seven original nodes in the following order and connect consecutive nodes by the six edges  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$ . The original topological order is therefore unique and directly checkable. Let one memory have capacity four; buffer  $a$  is a two-byte backed input and  $b$  a three-byte generated intermediate. The original order is

ALLOC( $a$ ), COPYIN( $a$ ), ALLOC( $b$ ), PRODUCE( $b$ ), FREE( $b$ ), USE( $a$ ), FREE( $a$ ).

There is no mandatory access to  $a$  between COPY-IN and USE. At  $b$ 's allocation, logical live volume is five and best-fit cannot find a three-byte hole. Evicting  $a$  lets  $b$  occupy  $[0, 3)$ . Because FREE is mandatory,  $b$  remains resident through its FREE;  $a$  is then reloaded before USE. Interrupting backed  $a$  charges only two bytes.

**Table S3:** Capacity-four pedagogical graph, from original order to augmented physical artifact.  $o_a, i_a$  are synthetic spill-out/spill-in nodes.

| Step | Executed event | Resident after event | Free intervals | Cumulative $E$ |
|------|----------------|----------------------|----------------|----------------|
| 1    | ALLOC( $a$ )   | $a : [0, 2)$         | $[2, 4)$       | 0              |
| 2    | COPYIN( $a$ )  | $a : [0, 2)$         | $[2, 4)$       | 0              |
| 3    | $o_a$          | none                 | $[0, 4)$       | 2              |
| 4    | ALLOC( $b$ )   | $b : [0, 3)$         | $[3, 4)$       | 2              |
| 5    | PRODUCE( $b$ ) | $b : [0, 3)$         | $[3, 4)$       | 2              |
| 6    | FREE( $b$ )    | none                 | $[0, 4)$       | 2              |
| 7    | $i_a$          | $a : [0, 2)$         | $[2, 4)$       | 2              |
| 8    | USE( $a$ )     | $a : [0, 2)$         | $[2, 4)$       | 2              |
| 9    | FREE( $a$ )    | none                 | $[0, 4)$       | 2              |

The spill table contains only  $(a, 0)$ , hence  $n_{\text{spill}} = 1, C = 2, D = 0, V = 2, E = 2$ . The validator also checks the original dependencies,  $\text{ALLOC}(a) \prec o_a \prec i_a \prec \text{FREE}(a)$ , the absence of an  $a$ -user between  $o_a$

and  $i_a$ , and range/nonoverlap of every half-open address interval. Interrupting a same-size generated buffer would charge four. This is Proposition 1’s unit-cost difference, not evidence that order generation automatically uses class labels.

### 3.3 End-to-end path on real Conv\_Case0

Printing all 2,580 original nodes is impractical, so Table S4 shows the complete method path by solver role. Scalable v2 considers at most 24 P2 members, each running ordering, best-fit, victim selection, spill insertion, and internal accounting/timing; the experimental harness then canonically validates the selected artifact. It has zero violations,  $E = 66,828$ , and 140 spills. Its traffic decomposes as  $C = 12,672$ ,  $D = 27,078$ ,  $V = 39,750$ , hence  $E = V + D = 39,750 + 27,078 = 66,828$ .

**Table S4:** Conv\_Case0 method path. The exact row proves traffic optimality only for fixed `unlock-frontier`.

| Role              | Artifact/search scope                   | $E$    | Spills | Supported conclusion                                    |
|-------------------|-----------------------------------------|--------|--------|---------------------------------------------------------|
| Predecessor       | iter038                                 | 88,044 | 339    | scalable P2 predecessor                                 |
| Production        | scalable v2, 24 candidates              | 66,828 | 140    | canonical-valid production result                       |
| Offline repair    | 10,000 stochastic single-node proposals | 65,532 | 137    | Conv0 case study, not a default stage                   |
| Fixed-order exact | frontier gap+packing                    | 57,408 | 112    | matching bound, packed, fixed-order traffic certificate |
| Official          | official P2 artifact                    | 73,500 | 141    | objective-matched comparator                            |

The exact row exposes the certificate chain. In the recorded 0.88-second run, gap CP-SAT reports both lower bound and objective 57,408. Contiguous packing is OPTIMAL (satisfaction only), standard schedule/memory/spill files are emitted, and the canonical evaluator returns zero violations and traffic 57,408. By Theorem 3 this is minimum *traffic* under `unlock-frontier`. It does not prove that 112 is the fewest spills among traffic ties and does not search every topological order. As controlled order-sensitivity evidence, the same planner fixed to the P1 order proves optimum traffic 81,504; this comparison is specific to Conv0.

### 3.4 Offline cost repair and the fixed-order oracle

The offline repair script observes expensive spill frontiers, proposes dependency-legal node or chain moves, replans the physical artifact, and accepts only strict P2-key improvements. Budgets are nonuniform: Conv\_Case0 uses 10,000 stochastic single-node hill proposals; Conv\_Case1 uses a width-10, two-round frontier beam; each FA instance receives 2,000 stochastic proposals with no accepted move; neither Matmul instance is searched. These studies only show that cost semantics may affect ordering in some cases. They are neither one uniform six-case algorithm nor a universal improvement.

The fixed-order oracle first solves weighted-gap selection in Eq. (4), then performs deterministic greedy packing trials and invokes an independent `NoOverlap2D` satisfaction model if needed. It emits the same schedule/memory/spill format as production and calls the same evaluator. On the eight internal 17-node oracle instances, production traffic under its *own selected fixed order* is, respectively, 352, 1,408, 704, 704, 352, 1,408, 704, and 704. Every value equals the fixed-order CP-SAT lower bound and validates. This is fixed-order agreement on eight discrete small graphs, not evidence of large-scale exactness or global optimality across orders.

### 3.5 Failures, unfinished cases, and reproduction boundary

A complete account includes failures rather than retaining only certificates:

- FA\_Case1 is a canonical-valid fixed-order FEASIBLE artifact with  $E = 32,512$ , lower bound 31,936, and gap 576; it is not certified optimal.
- The reproducible 120.09-second Matmul\_Case0 run is FEASIBLE at 34,816 with lower bound 29,952. A legal heuristic artifact on the same order reaches 34,688, so the audited interval is  $[29,952, 34,688]$ ; the old “34,688 OPTIMAL” claim is revoked.

- Conv\_Case1 did not finish exact+packing within the research budget; Matmul\_Case1 was not run.
- Against official P3 time, the six public cases yield five wins and one loss. Conv\_Case1 is 1,118,687 versus 1,073,322, so dominance is not universal.
- Summary JSON can check metrics and validation status, but independent replay requires rerunning the deterministic solver when matching text artifacts are absent. Fixed-order certificates instead use machine JSON plus matching schedule/memory/spill files; older handwritten status cannot override machine records.

Address placement and residency scans make the production path scalable for a fixed portfolio. CP-SAT gap selection and fallback packing are exponential in the worst case. Canonical solve currently has no CP-SAT size threshold, timeout switch, or fallback integration, so oracle successes are not capabilities of the default solver.

## 4 Complete Evidence and Negative Results

This section expands the compact evaluation in the main paper and states the objective and artifact role of every table. P2 and P3 come from separate searches with different lexicographic keys; exact results are research-oracle evidence, not production outputs.

### 4.1 Complete public P1, P2, and P3 results

The six public instances are the released DAGs of Problem A from the 2025 Huawei Cup China Graduate Mathematical Contest in Modeling; the synthetic families reported later are separate internal test sets.

Table S5 completes the current solver’s P1 logical-lifetime metrics. It is regenerated from `results/autoresearch_v2/round12_audited_p1.json`; all six rows have zero evaluator violations. Parentheses contain the matching official P1 artifact. L1/UB are ALLOC-FREE prefix-sum peaks and L0A/B/C are counts of simultaneously logically live buffers, not post-spill P2 physical-residency traces. The current order slightly lowers L1 on both Conv instances but greatly increases live L0B counts. This negative result prevents collapsing P1 components and time into a claim of universal superiority, and P1 cannot replace the P2/P3 artifact comparison.

**Table S5:** P1 logical metrics on the six public instances, reported as ours (official). Time is evaluator-computed schedule cycles.

| Instance     | L1              | UB             | L0A     | L0B       | L0C    | Time              |
|--------------|-----------------|----------------|---------|-----------|--------|-------------------|
| Conv_Case0   | 6,912 (7,488)   | 0 (0)          | 0 (0)   | 300 (1)   | 0 (0)  | 394,430 (359,570) |
| Conv_Case1   | 13,786 (14,040) | 0 (0)          | 0 (0)   | 5,264 (1) | 0 (0)  | 650,044 (573,395) |
| FA_Case0     | 256 (3,328)     | 8,258 (1,428)  | 24 (1)  | 32 (1)    | 4 (1)  | 31,063 (31,429)   |
| FA_Case1     | 256 (5,376)     | 33,010 (2,468) | 112 (1) | 128 (1)   | 8 (1)  | 112,458 (113,445) |
| Matmul_Case0 | 128 (9,216)     | 0 (0)          | 57 (1)  | 448 (1)   | 8 (1)  | 82,308 (82,773)   |
| Matmul_Case1 | 128 (34,816)    | 0 (0)          | 241 (1) | 3,840 (1) | 16 (1) | 629,076 (583,525) |

P2 has five strict wins and one tie; its median reduction is 0.866% and maximum 9.08%. Matmul\_Case1 ties traffic but wins the time tie-break. P3 has five time wins and one loss, with median reduction 3.77%; Conv\_Case1 regresses by 4.23%. These statistics compare production only with objective-matched official P2 or P3 artifacts and exclude repair and the oracle. The JSON files are metric/validation ledgers, not colocated archived schedule/memory/spill triplets; independent replay requires rerunning the deterministic solver. Wall times document these runs and support neither stable throughput nor a scaling law.

**Table S6:** Complete production P2 records. Vol is raw spill volume and Dt generated-buffer volume. Wall is one audited solver run, not a repeated performance estimate. All rows are canonical-valid with zero violations.

| Instance     | Tr      | Spills | Vol     | Dt     | $T$       | Sched. len. | Official Tr | Result | Wall (s) |
|--------------|---------|--------|---------|--------|-----------|-------------|-------------|--------|----------|
| Conv_Case0   | 66,828  | 140    | 39,750  | 27,078 | 521,034   | 2,860       | 73,500      | win    | 0.552    |
| Conv_Case1   | 72,734  | 1,987  | 69,731  | 3,003  | 1,154,672 | 40,060      | 73,240      | win    | 43.055   |
| FA_Case0     | 3,584   | 14     | 1,792   | 1,792  | 44,956    | 1,744       | 3,692       | win    | 0.263    |
| FA_Case1     | 32,512  | 127    | 20,480  | 12,032 | 164,166   | 7,206       | 32,840      | win    | 2.216    |
| Matmul_Case0 | 34,688  | 271    | 34,688  | 0      | 192,380   | 4,702       | 34,944      | win    | 0.927    |
| Matmul_Case1 | 460,800 | 3,600  | 460,800 | 0      | 1,783,083 | 38,176      | 460,800     | tie    | 20.973   |

**Table S7:** Complete production P3 records. P3 is time-first; official traffic and time come from matching P3 artifacts. Wall again records only one audited run. All rows are canonical-valid with zero violations.

| Instance     | Tr      | Spills | $T$       | Sched. len. | Official Tr | Official $T$ | Time result | Wall (s) |
|--------------|---------|--------|-----------|-------------|-------------|--------------|-------------|----------|
| Conv_Case0   | 69,712  | 147    | 515,634   | 2,874       | 73,500      | 535,312      | win         | 1.106    |
| Conv_Case1   | 72,742  | 2,015  | 1,118,687 | 40,116      | 86,690      | 1,073,322    | loss        | 61.133   |
| FA_Case0     | 4,096   | 16     | 36,344    | 1,748       | 4,128       | 46,761       | win         | 0.434    |
| FA_Case1     | 34,304  | 134    | 152,364   | 7,220       | 32,840      | 193,059      | win         | 3.119    |
| Matmul_Case0 | 34,944  | 273    | 186,820   | 4,706       | 34,944      | 194,331      | win         | 1.585    |
| Matmul_Case1 | 460,928 | 3,601  | 1,771,383 | 38,178      | 460,800     | 1,800,218    | win         | 36.963   |

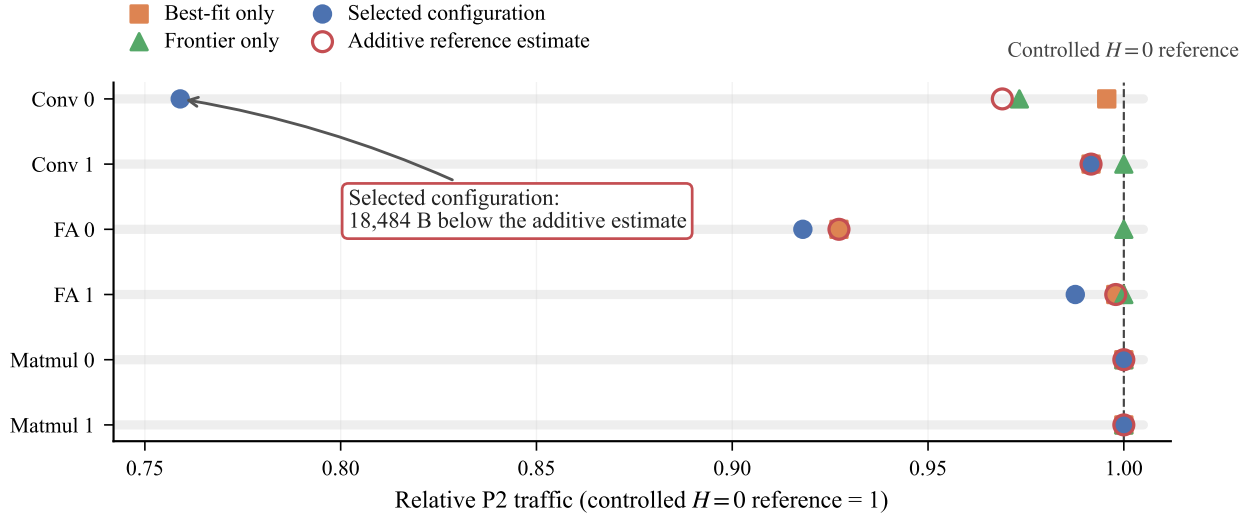
#### 4.2 Selected ablation path and descriptive nonadditivity

Table S8 reports selected configurations from a controlled  $H = 0$  ablation. The reference is not the actual promoted predecessor, and the final  $H = 0$  configuration is not the complete canonical portfolio. Every round-7 configuration selects by  $(E, n)$  only. The design is also not factorial: two pairwise combinations are absent. It can reveal a nonadditive measured path, but it cannot identify isolated main effects or an interaction coefficient.

**Table S8:** Selected  $H = 0$  traffic-ablation configurations. BF denotes best-fit and F frontier ordering. “Selected” also enables the adaptive policy; it is a measured configuration, not a factorial estimate.

| Case         | Actual pred. | $H = 0$ ref. | BF only | F only  | Selected | $\Delta$ prod. |
|--------------|--------------|--------------|---------|---------|----------|----------------|
| Conv_Case0   | 88,044       | 88,044       | 87,660  | 85,696  | 66,828   | −21, 216       |
| Conv_Case1   | 72,520       | 73,348       | 72,734  | 73,348  | 72,734   | +214           |
| FA_Case0     | 3,904        | 3,904        | 3,620   | 3,904   | 3,584    | −320           |
| FA_Case1     | 32,920       | 32,920       | 32,852  | 32,920  | 32,512   | −408           |
| Matmul_Case0 | 34,688       | 34,688       | 34,688  | 34,688  | 34,688   | 0              |
| Matmul_Case1 | 460,800      | 460,800      | 460,800 | 460,800 | 460,800  | 0              |

For Conv\_Case0, the naive sum of the measured best-fit-only and frontier-only deltas predicts 85,312 bytes, while the selected all-mechanism  $H = 0$  configuration reaches 66,828, a descriptive deviation of 18,484 bytes. This gap is evidence that the measured configurations are nonadditive, not a claim of a separately identified synergy: the adaptive policy is present in the selected configuration and the missing pairwise cells prevent factorial attribution. FA\_Case0 likewise moves from 4,608 under frontier+adaptive with first-fit to 3,584 when best-fit is added, showing that the policy’s outcome depends on placement. Conv\_Case1 is a further boundary: the selected result improves over the controlled  $H = 0$  reference but is 214 bytes worse than the actual promoted predecessor. The Matmul cases are invariant. These rows support complete-configuration evaluation,



**Figure S1:** Relative P2 traffic for selected controlled  $H = 0$  configurations. The open circle is a reference estimate obtained by directly adding the measured best-fit-only and frontier-only changes. It visualizes nonadditivity along this configuration path but is not a factorial attribution. Against the actual predecessor, the selected configuration has three wins, two ties, and one loss.

not a universal per-module benefit.

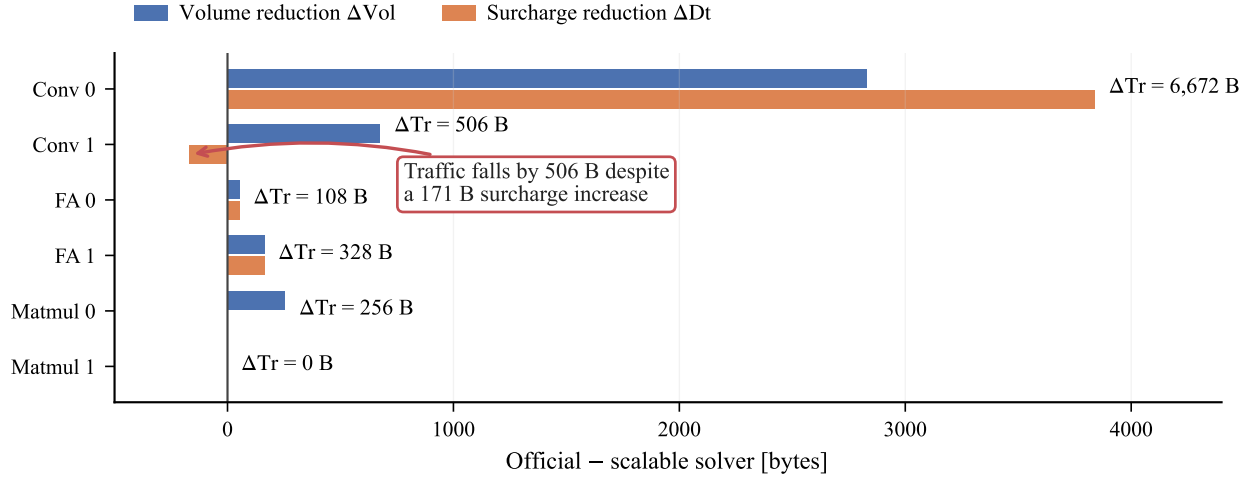
#### 4.3 Three spill-cost regimes under $\text{Tr}=\text{Vol}+\text{Dt}$

Table S9 pairs every scalable P2 output with its official P2 artifact. We report total raw spill volume Vol and its generated/unbacked part Dt, so each row obeys  $\text{Tr} = \text{Vol} + \text{Dt}$ . Deltas are official minus scalable; hence  $\Delta\text{Tr} = \Delta\text{Vol} + \Delta\text{Dt}$ . This paired accounting distinguishes volume reduction from surcharge reduction without assuming either mechanism.

**Table S9:** Paired all-case P2 decomposition in bytes. Deltas are official minus scalable and satisfy  $\Delta\text{Tr} = \Delta\text{Vol} + \Delta\text{Dt}$ .

| Case         | Off. Vol | Off. Dt | Off. Tr | Ours Vol | Ours Dt | Ours Tr | $\Delta\text{Vol}$ | $\Delta\text{Dt}$ | $\Delta\text{Tr}$ |
|--------------|----------|---------|---------|----------|---------|---------|--------------------|-------------------|-------------------|
| Conv_Case0   | 42,582   | 30,918  | 73,500  | 39,750   | 27,078  | 66,828  | 2,832              | 3,840             | 6,672             |
| Conv_Case1   | 70,408   | 2,832   | 73,240  | 69,731   | 3,003   | 72,734  | 677                | -171              | 506               |
| FA_Case0     | 1,846    | 1,846   | 3,692   | 1,792    | 1,792   | 3,584   | 54                 | 54                | 108               |
| FA_Case1     | 20,644   | 12,196  | 32,840  | 20,480   | 12,032  | 32,512  | 164                | 164               | 328               |
| Matmul_Case0 | 34,944   | 0       | 34,944  | 34,688   | 0       | 34,688  | 256                | 0                 | 256               |
| Matmul_Case1 | 460,800  | 0       | 460,800 | 460,800  | 0       | 460,800 | 0                  | 0                 | 0                 |

The rows span three regimes. FA\_Case0 is generated-only ( $\text{Dt} = \text{Vol}$ ), both Matmul cases are backed-only ( $\text{Dt} = 0$ ), and the Conv and FA\_Case1 rows are mixed. All five strict P2 wins reduce Vol. The surcharge Dt also falls for Conv\_Case0, FA\_Case0, and FA\_Case1, remains zero for Matmul\_Case0, but *increases* by 171 bytes for Conv\_Case1. Conv\_Case1 still wins because its 677-byte volume reduction outweighs that surcharge regression, giving  $506 = 677 - 171$ . FA\_Case0 reduces Vol and Dt by the same 54 bytes, so its class share does not change; Matmul\_Case0 wins with  $\text{Dt} = 0$  throughout. These paired facts directly rule out one composition story for all five wins.



**Figure S2:** Per-instance P2 accounting relative to the official artifact. Every strict win reduces total spill volume Vol, whereas the generated surcharge Dt may rise or fall. Their signed changes sum to the net traffic reduction,  $\Delta\text{Tr} = \Delta\text{Vol} + \Delta\text{Dt}$ .

#### 4.4 Within-portfolio P2-to-P3 tradeoff

P2 selects by  $(\text{Tr}, n, T)$ , whereas P3 selects by  $(T, \text{Tr}, n)$ . Table S10 compares the two artifacts produced by our own bounded portfolio on the same case. This controls the solver and instance, but it does not establish a global Pareto frontier.

**Table S10:** Within-portfolio objective tradeoff.  $\text{Tr}_2, T_2$  belong to the P2-selected artifact and  $\text{Tr}_3, T_3$  to the independently selected P3 artifact.

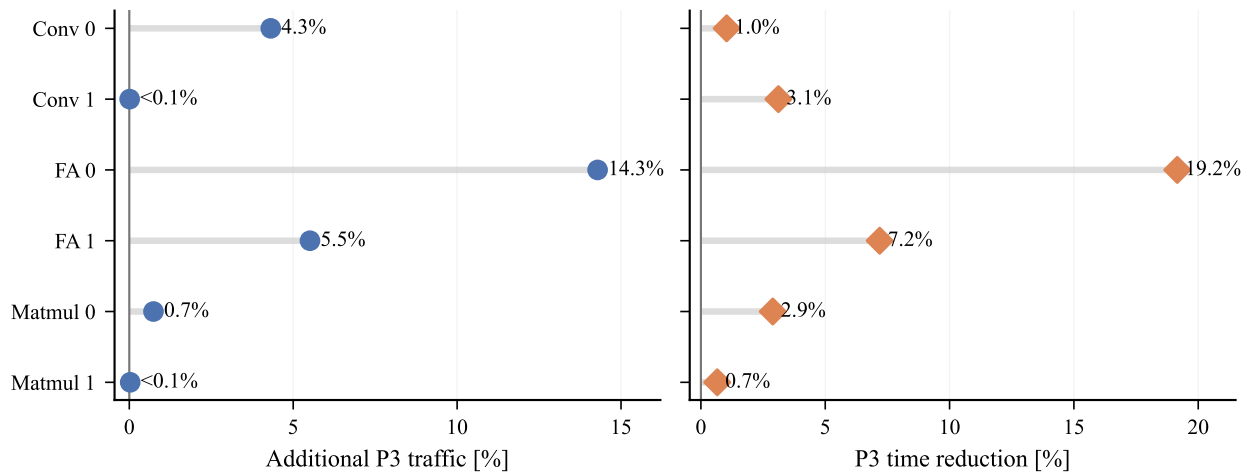
| Case         | $\text{Tr}_2$ | $\text{Tr}_3$ | $\text{Tr}_3 - \text{Tr}_2$ | $T_2$     | $T_3$     | $T_3 - T_2$ |
|--------------|---------------|---------------|-----------------------------|-----------|-----------|-------------|
| Conv_Case0   | 66,828        | 69,712        | 2,884                       | 521,034   | 515,634   | −5,400      |
| Conv_Case1   | 72,734        | 72,742        | 8                           | 1,154,672 | 1,118,687 | −35,985     |
| FA_Case0     | 3,584         | 4,096         | 512                         | 44,956    | 36,344    | −8,612      |
| FA_Case1     | 32,512        | 34,304        | 1,792                       | 164,166   | 152,364   | −11,802     |
| Matmul_Case0 | 34,688        | 34,944        | 256                         | 192,380   | 186,820   | −5,560      |
| Matmul_Case1 | 460,800       | 460,928       | 128                         | 1,783,083 | 1,771,383 | −11,700     |

The P3 search accepts more traffic than the P2 output on every public case, from 8 bytes on Conv\_Case1 to 2,884 on Conv\_Case0, and it returns a faster artifact than the P2 search on every case. Conv\_Case1 is the sharpest near-zero-traffic example: 8 additional bytes accompany 35,985 fewer cycles. This is the direct within-portfolio evidence for objective decoupling. The official P2 and P3 artifacts are different baselines, so the headline fact that Conv\_Case1 wins official P2 but loses official P3 is only a warning not to transfer a win across objectives; it is not used to infer this tradeoff.

#### 4.5 Case-study repair and fixed-order research oracle

The Conv\_Case0 and Conv\_Case1 gains are therefore evidence from two repair prototypes, not a uniform algorithm evaluated at a common budget. Separate 2,000-proposal, seed-0 hill probes on FA\_Case0 and FA\_Case1 accepted no move; that smaller negative probe is not comparable to either Conv budget. No Matmul repair search was run; the reported Matmul rows elsewhere simply reuse the structural order.

Conv\_Case0/unlock-frontier validates with 112 spills in 0.88 s. The same planner also certifies the legacy P1 order at 81,504 with 328 spills in 19.49 s. Their 24,096-byte gap on the same case isolates a fixed-order



**Figure S3:** Within-portfolio change from the P2-selected to the independently selected P3 artifact. The aligned panels separate the additional traffic from the time reduction. Every observed P3 artifact trades more traffic for less time; this is a bounded-portfolio observation, not a global Pareto claim.

**Table S11:** Two nonuniform cost-aware order-repair case studies. The search procedures and budgets differ and do not define one six-case method stage.

| Case       | Base Tr | Repair Tr | $\Delta$ Tr | Search budget                            |
|------------|---------|-----------|-------------|------------------------------------------|
| Conv_Case0 | 66,828  | 65,532    | −1, 296     | 10,000-proposal single-node hill, seed 0 |
| Conv_Case1 | 72,734  | 70,940    | −1, 794     | spill-frontier beam 10, two rounds       |

**Table S12:** Fixed-order research-oracle traffic evidence. “Certificate” means machine-readable gap result, packed schedule/memory/spill artifacts, canonical validation, and traffic equal to the lower bound.

| Case         | Fixed order     | Traffic | LB     | Gap   | Evidence status            |
|--------------|-----------------|---------|--------|-------|----------------------------|
| Conv_Case0   | unlock-frontier | 57,408  | 57,408 | 0     | traffic certificate        |
| Conv_Case0   | legacy P1       | 81,504  | 81,504 | 0     | traffic certificate        |
| FA_Case0     | id-raw          | 3,584   | 3,584  | 0     | traffic certificate        |
| FA_Case1     | capfit-id       | 32,512  | 31,936 | 576   | machine-checkable feasible |
| Matmul_Case0 | capfit-id       | 34,816  | 29,952 | 4,864 | machine-checkable feasible |
| Conv_Case1   | —               | —       | —      | —     | run-log packing timeout    |
| Matmul_Case1 | —               | —       | —      | —     | not run                    |

bottleneck without claiming global order optimality. FA\_Case0 is the third machine-checkable traffic certificate and extends coverage to a second public instance (14 spills, 2.17 s). FA\_Case1 and Matmul\_Case0 have canonical-valid packed artifacts, but their positive lower-bound gaps make them feasible/provisional rather than traffic-optimal. The old Matmul\_Case0 claim of 34,688 optimality is withdrawn. CP-SAT remains a research oracle; canonical solve does not invoke it, and none of these certificates optimizes spill count or time among equal-traffic plans.

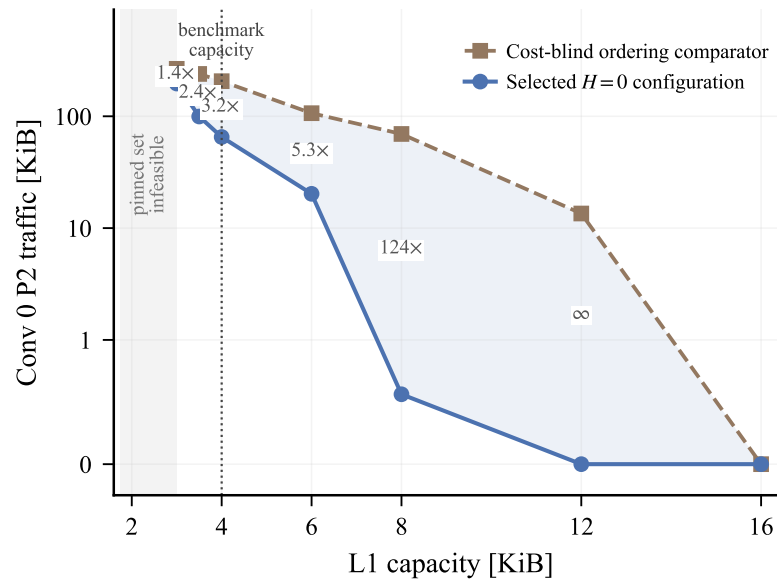
The aligned residency traces and their certificate values are promoted to the main paper because they directly support the order-space mechanism claim.

#### 4.6 Single-instance capacity boundary

**Table S13:** Conv\_Case0 L1-capacity sweep. Infeasible rows exceed capacity at a mandatory pinned event and are not algorithm losses.

| L1 capacity | Selected Tr | Cost-blind order Tr | Ratio   | Selected order    |
|-------------|-------------|---------------------|---------|-------------------|
| 2,048       | –           | –                   | –       | pinned infeasible |
| 2,560       | –           | –                   | –       | pinned infeasible |
| 3,072       | 200,960     | 272,010             | 1.354   | capfit-id         |
| 3,584       | 101,924     | 244,868             | 2.402   | unlock-frontier   |
| 4,096       | 66,828      | 210,876             | 3.156   | unlock-frontier   |
| 6,144       | 20,744      | 109,116             | 5.260   | unlock-frontier   |
| 8,192       | 576         | 71,144              | 123.514 | unlock-frontier   |
| 12,288      | 0           | 13,830              | –       | unlock-frontier   |
| 16,384      | 0           | 0                   | –       | unlock-frontier   |

The controlled fixed- $H = 0$  selection never loses the cost-blind ordering comparator at a feasible capacity in this sweep. It evaluates four structural orders, two victim policies, and best-fit placement. The comparator fixes the cost-blind cp-free-first order but still optimizes best-fit placement and both victim policies, including the asymmetric policy; it is not a wholly cost-unaware planner. The selected order is capfit-id at 3,072 and unlock-frontier from 3,584 onward. Ratios become unstable near zero traffic, so Table S13 retains raw values and leaves zero-denominator ratios blank. The sweep covers only Conv\_Case0, L1, fixed  $H = 0$ , and traffic. Its records contain no per-point  $C/D$ , spill count, time, or canonical-validation field, so we do not extrapolate it to cross-operator or cross-cache robustness.



**Figure S4:** Controlled Conv\_Case0 L1-capacity sweep at fixed  $H = 0$  (symlog vertical axis). Labels between the curves give the traffic ratio between the comparator and the selected configuration; the shaded band marks the gap due to topological-order choice in this controlled slice. Capacities below 3,072 bytes are infeasible because the mandatory pinned set itself exceeds capacity.

#### 4.7 Standard validated re-evaluation on internal synthetic benchmarks

The earlier fixed- $H = 0$  assign/extra-split diagnostic lacked canonical validation and is superseded here; it is no longer a paper result. We reran all three internal sets without changing an instance: 36 graphs (UB, L0C, and L1 targets; 18 capacity-bound and 18 order-reachable), eight 17-node oracle instances, and a clean/dirty mirror pair. Each of the 36 graphs evaluates seven configurations, for 252 runs. The script generates each schedule/address/spill artifact in memory and immediately calls the canonical evaluator. All 252 CSV rows record `valid=True` and zero violations. The metric files are regenerated by `scripts/paper/v2_synth_suite.py`.

**Table S14:** Canonically validated comparison of the production solver on the 36-instance synthetic suite. The predecessor portfolio is the full portfolio selection over the `capfit-id/P1/id-row` candidate orders with the distance-based victim policy; the four cost-blind order baselines follow the original suite definitions; random-best takes the best of eight seeded random topological orders. The median ratio is our Tr over the comparator’s Tr (with 0/0 counted as 1).

| Comparator            | All 36 W/T/L | Capacity-bound 18 | Order-reachable 18 | Median ratio |
|-----------------------|--------------|-------------------|--------------------|--------------|
| Predecessor portfolio | 0/36/0       | 0/18/0            | 0/18/0             | 1.00         |
| cp-list               | 30/6/0       | 18/0/0            | 12/6/0             | 0.39         |
| cp-free-first         | 14/22/0      | 14/4/0            | 0/18/0             | 1.00         |
| pressure-uniform      | 13/23/0      | 10/8/0            | 3/15/0             | 1.00         |
| goodman-hsu           | 19/17/0      | 7/11/0            | 12/6/0             | 0.67         |
| random-best           | 27/9/0       | 18/0/0            | 9/9/0              | 0.34         |

Three conclusions follow. First, production ties the predecessor on all 36 graphs. Since predecessor configurations are a subset of production candidates, this zero-loss relation follows from portfolio monotonicity and is not independent generalization evidence; the new module adds no gain here. Second, `cp-free-first` and production’s `id-row` order are identical on 36/36 graphs, and its distance policy at  $H = 0$  also lies in the production search space. The 14 wins/22 ties against it are therefore another inclusion-based non-regression check. The nonincluded comparisons are `cp-list`, `pressure-uniform`, `Goodman-Hsu`, and `best-of-8` random orders. Production has zero losses against these four groups only under the recorded implementations and budgets. Third, all 252 runs validate with zero violations.

Tables S15–S16 give every per-instance record; random denotes best-of-8.

**Table S15:** All 36 instances, part I: properties, production result, and predecessor. C/R denotes capacity-bound/order-reachable; numeric results are bytes.

| Instance  | Regime | Cache | Nodes | Ours Tr | $C$   | $D$   | Spills | Pred. Tr |
|-----------|--------|-------|-------|---------|-------|-------|--------|----------|
| synth_000 | C      | L0C   | 35    | 224     | 224   | 0     | 2      | 224      |
| synth_001 | R      | L1    | 38    | 0       | 0     | 0     | 0      | 0        |
| synth_002 | C      | UB    | 40    | 1,088   | 0     | 544   | 2      | 1,088    |
| synth_003 | R      | L0C   | 40    | 0       | 0     | 0     | 0      | 0        |
| synth_004 | C      | L1    | 34    | 1,376   | 1,376 | 0     | 2      | 1,376    |
| synth_005 | R      | UB    | 30    | 0       | 0     | 0     | 0      | 0        |
| synth_006 | C      | L0C   | 39    | 448     | 0     | 224   | 2      | 448      |
| synth_007 | R      | L1    | 32    | 0       | 0     | 0     | 0      | 0        |
| synth_008 | C      | UB    | 44    | 544     | 544   | 0     | 2      | 544      |
| synth_009 | R      | L0C   | 46    | 0       | 0     | 0     | 0      | 0        |
| synth_010 | C      | L1    | 38    | 2,752   | 0     | 1,376 | 2      | 2,752    |
| synth_011 | R      | UB    | 24    | 0       | 0     | 0     | 0      | 0        |
| synth_012 | C      | L0C   | 43    | 224     | 224   | 0     | 2      | 224      |
| synth_013 | R      | L1    | 38    | 0       | 0     | 0     | 0      | 0        |

Continued on next page

Table S15 (continued)

| Instance  | Regime | Cache | Nodes | Ours Tr | $C$   | $D$   | Spills | Pred. Tr |
|-----------|--------|-------|-------|---------|-------|-------|--------|----------|
| synth_014 | C      | UB    | 28    | 1,088   | 0     | 544   | 2      | 1,088    |
| synth_015 | R      | L0C   | 40    | 0       | 0     | 0     | 0      | 0        |
| synth_016 | C      | L1    | 42    | 1,376   | 1,376 | 0     | 2      | 1,376    |
| synth_017 | R      | UB    | 30    | 0       | 0     | 0     | 0      | 0        |
| synth_018 | C      | L0C   | 47    | 448     | 0     | 224   | 2      | 448      |
| synth_019 | R      | L1    | 32    | 0       | 0     | 0     | 0      | 0        |
| synth_020 | C      | UB    | 32    | 544     | 544   | 0     | 2      | 544      |
| synth_021 | R      | L0C   | 46    | 0       | 0     | 0     | 0      | 0        |
| synth_022 | C      | L1    | 46    | 2,752   | 0     | 1,376 | 2      | 2,752    |
| synth_023 | R      | UB    | 24    | 0       | 0     | 0     | 0      | 0        |
| synth_024 | C      | L0C   | 31    | 224     | 224   | 0     | 2      | 224      |
| synth_025 | R      | L1    | 38    | 0       | 0     | 0     | 0      | 0        |
| synth_026 | C      | UB    | 36    | 1,088   | 0     | 544   | 2      | 1,088    |
| synth_027 | R      | L0C   | 40    | 0       | 0     | 0     | 0      | 0        |
| synth_028 | C      | L1    | 50    | 1,376   | 1,376 | 0     | 2      | 1,376    |
| synth_029 | R      | UB    | 30    | 0       | 0     | 0     | 0      | 0        |
| synth_030 | C      | L0C   | 35    | 448     | 0     | 224   | 2      | 448      |
| synth_031 | R      | L1    | 32    | 0       | 0     | 0     | 0      | 0        |
| synth_032 | C      | UB    | 40    | 544     | 544   | 0     | 2      | 544      |
| synth_033 | R      | L0C   | 46    | 0       | 0     | 0     | 0      | 0        |
| synth_034 | C      | L1    | 34    | 2,752   | 0     | 1,376 | 2      | 2,752    |
| synth_035 | R      | UB    | 24    | 0       | 0     | 0     | 0      | 0        |

**Table S16:** All 36 instances, part II: Tr of five comparison orders. Random is best-of-8.

| Instance  | cp-list | cp-free-first | pressure | Goodman-Hsu | random |
|-----------|---------|---------------|----------|-------------|--------|
| synth_000 | 560     | 336           | 224      | 224         | 688    |
| synth_001 | 1,024   | 0             | 0        | 256         | 0      |
| synth_002 | 2,464   | 2,176         | 1,520    | 1,632       | 3,984  |
| synth_003 | 384     | 0             | 192      | 48          | 336    |
| synth_004 | 2,064   | 1,376         | 1,376    | 2,064       | 2,064  |
| synth_005 | 0       | 0             | 0        | 0           | 0      |
| synth_006 | 1,184   | 896           | 800      | 448         | 1,152  |
| synth_007 | 1,376   | 0             | 0        | 688         | 688    |
| synth_008 | 1,936   | 1,360         | 1,104    | 1,088       | 2,896  |
| synth_009 | 256     | 0             | 0        | 32          | 160    |
| synth_010 | 5,504   | 2,752         | 2,752    | 2,752       | 5,856  |
| synth_011 | 0       | 0             | 0        | 0           | 0      |
| synth_012 | 688     | 448           | 336      | 336         | 1,024  |
| synth_013 | 1,024   | 0             | 0        | 256         | 0      |
| synth_014 | 1,632   | 1,632         | 1,088    | 1,088       | 1,664  |
| synth_015 | 384     | 0             | 192      | 48          | 336    |
| synth_016 | 3,440   | 2,064         | 1,376    | 1,376       | 4,480  |
| synth_017 | 0       | 0             | 0        | 0           | 0      |
| synth_018 | 1,312   | 896           | 864      | 672         | 1,696  |
| synth_019 | 1,376   | 0             | 0        | 688         | 688    |
| synth_020 | 1,088   | 1,088         | 816      | 544         | 1,504  |
| synth_021 | 256     | 0             | 0        | 32          | 160    |
| synth_022 | 8,256   | 5,504         | 3,104    | 2,752       | 7,936  |
| synth_023 | 0       | 0             | 0        | 0           | 0      |
| synth_024 | 448     | 224           | 336      | 224         | 448    |
| synth_025 | 1,024   | 0             | 0        | 256         | 0      |

Continued on next page

Table S16 (continued)

| Instance  | cp-list | cp-free-first | pressure | Goodman–Hsu | random |
|-----------|---------|---------------|----------|-------------|--------|
| synth_026 | 2,320   | 2,176         | 1,376    | 1,088       | 3,440  |
| synth_027 | 384     | 0             | 192      | 48          | 336    |
| synth_028 | 4,480   | 2,752         | 1,376    | 1,376       | 5,872  |
| synth_029 | 0       | 0             | 0        | 0           | 0      |
| synth_030 | 1,120   | 672           | 448      | 448         | 1,152  |
| synth_031 | 1,376   | 0             | 0        | 688         | 688    |
| synth_032 | 1,376   | 1,088         | 960      | 816         | 2,048  |
| synth_033 | 256     | 0             | 0        | 32          | 160    |
| synth_034 | 4,128   | 2,752         | 2,752    | 4,128       | 4,128  |
| synth_035 | 0       | 0             | 0        | 0           | 0      |

**Table S17:** Production solver versus the fixed-order CP-SAT traffic optimum computed on the solver’s own selected topological order, on the 8-instance oracle set. All eight solver outputs equal the optimum exactly (zero gap, status OPTIMAL in every row).

| Instance   | Bound cache | Ours Tr | Fixed-order optimum | Gap |
|------------|-------------|---------|---------------------|-----|
| oracle_000 | L0C         | 352     | 352                 | 0   |
| oracle_001 | UB          | 1,408   | 1,408               | 0   |
| oracle_002 | L0C         | 704     | 704                 | 0   |
| oracle_003 | UB          | 704     | 704                 | 0   |
| oracle_004 | L0C         | 352     | 352                 | 0   |
| oracle_005 | UB          | 1,408   | 1,408               | 0   |
| oracle_006 | L0C         | 704     | 704                 | 0   |
| oracle_007 | UB          | 704     | 704                 | 0   |

The oracle set provides a quantitative yardstick that the public instances cannot deliver in full: for each 17-node instance we ran the fixed-order CP-SAT model of Section 2 on the topological order the production solver actually selected, and all eight instances solved to optimality with objectives exactly equal to the solver’s output (Table S17). On this small family, the scalable eviction heuristic therefore leaves no traffic headroom under its own topological order. The claim covers only 17-node instances and the traffic objective, and does not extrapolate to public scale: the 14.1% gap between the production result 66,828 and the same-order certificate 57,408 on Conv\_Case0 (main text, Section 4) shows that headroom does exist on large instances.

The clean/dirty mirror pair re-checks the  $\text{Tr} = \text{Vol} + \text{Dt}$  billing semantics on a minimal example: both instances spill the same volume  $\text{Vol} = 1,024$  with the same spill count of two, yet the clean variant is fully backed ( $\text{Dt} = 0$ ,  $\text{Tr} = 1,024$ ) while the dirty variant is fully generated ( $\text{Dt} = 1,024$ ,  $\text{Tr} = 2,048$ ), doubling the billed traffic exactly; the production solver and the predecessor portfolio tie on both.